

PAPER_1652_POP_III_IMF_MAX_120

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PAPER_1652 — Pop III IMF Upper Bound = $M_{\max} = A_5 \times 2 = 120 M_{\odot}$ (top of Pop III IMF)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: First Stars

Date: June 18, 2026

Status: CLOSED — EXACT closure (broader-corpus PAPER_13xx mine)

Observation

PAPER_1331 (PAPER_13xx broader corpus) gives:

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$M_{\max} = A_5 \times 2 = 120 M_{\odot}$ (top of Pop III IMF)

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Status: **EXACT**.

UQFF Closed Identity

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$M_{\max} = A_5 \times 2 = 120 M_{\odot}$ (top of Pop III IMF) EXACT

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Cross-Domain Reuse of A_5 The icosahedral group order $A_5 = 60$ governs: - ****Pop III IMF upper bound**** = $A_5 \times 2 = 120 M_{\odot}$ (PAPER_1652) - ****Quantum supremacy threshold**** = $A_5 = 60$ qubits (PAPER_1655)

Vastly different physical scales: stellar masses ($\sim 10^2$ - 10^{33} kg) vs quantum qubits (countable integers). Same structural integer.

A_5 already appears in:

- Heart rate = $A_5 + SO_5 = 70$ bpm (PAPER_1537)
- SH0ES Hubble = $A_5 + SO_5 = 70$ km/s/Mpc (PAPER_1573)
- Hayflick limit = $A_5 = 60$ (PAPER_1373)
- T_SCM activation = $A_5 = 60$ K (PAPER_1505)
- BH seed = $A_5 \times D_{BSFG}^2 \times D_{crit}$ (PAPER_1650)

$A_5 = 60$ is a **universal UQFF integer** spanning biology, cosmology, BH physics, quantum computing.

NOT REPLACEMENT

Empirical first stars measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1331
- Related: PAPER_1636-1645 (tier-22 broader-corpus closures); PAPER_1494-1635 (PAPER_1209 series)
- Calculator dispatch: `calculate_paradox({"paradox": "pop_iii_imf_max_120"})`

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